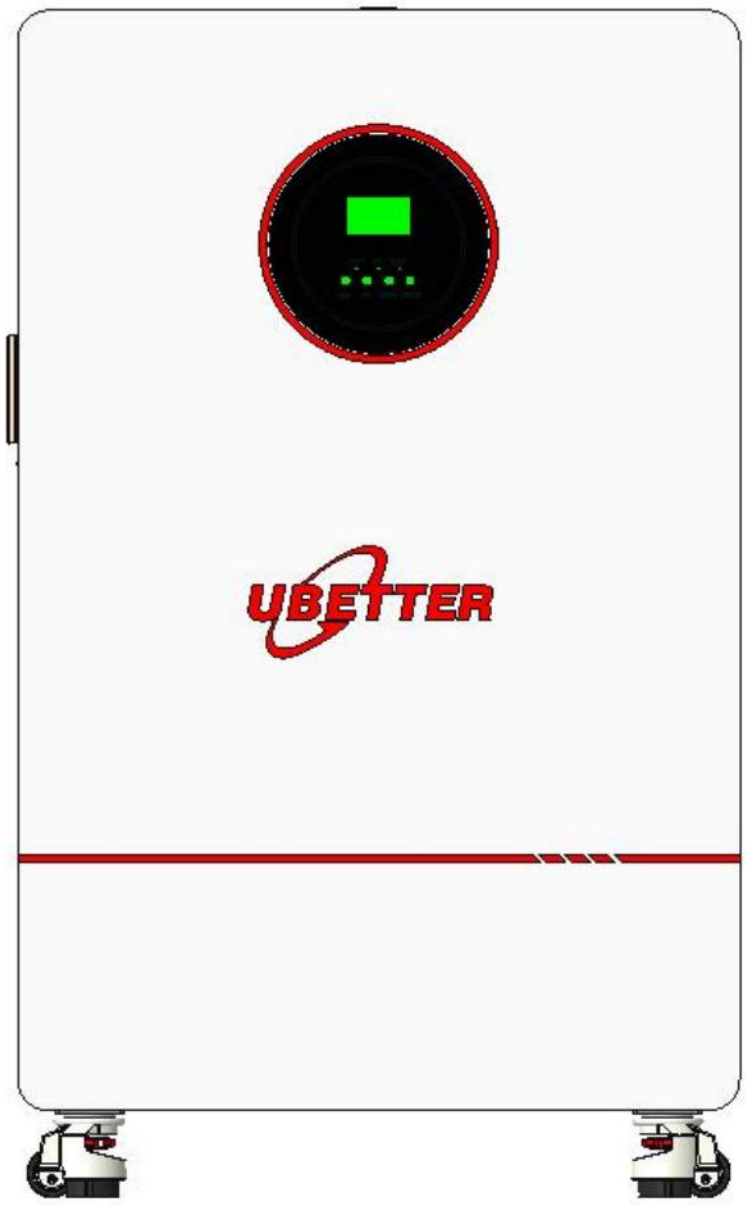


UBETTER TECHNOLOGY Co.,Ltd

5KW+15kWh inverter all-in-one machine

specification

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1. scope

This specification applies to the 48V 314Ah 15S1P LiFePO4 rechargeable battery pack manufactured by UBETTER TECHNOLOGY COMPANY LIMITED.

2. main specifications

2.1 Battery cell specification TED

	No.	project	Common parameters	remarks
Cell	1	nominal capacity	314Ah	
	2	nominal voltage	3.2V	Average operating voltage
	3	Internal impedance	0.17±0.05 mΩ	AC 1 KHz test (50% SOC)
	4	size	(71.7±0.5) *(173.9±0.8) *(204.3±0.8)	Initial dimensions
	5	weight	5.60±0.15Kg	make an appointment
	6	Charging cut-off voltage	3.65 V	Cell temperature T> 0℃ Cell temperature T> 0℃
	7	Discharge cutoff voltage	2.5 V	Battery temperature T>0℃ Cell temperature T > 0℃
			2.0 V	Battery temperature T ≤ 0℃ Cell temperature T≤0℃
	8	Charging rate	Standard ratio: 0.5P	25 ℃
	9	Discharge rate	Standard ratio: 0.5P	25 ℃
			Maximum continuous discharge rate: 1P	25 ℃
	10	storage temperature	Temperature: -10℃~35℃, humidity ≤ 85%	
	11	Cyclic performance	25℃ 0.5C/0.5C 8000 times	Capacity retention rate is greater than or equal to 70%

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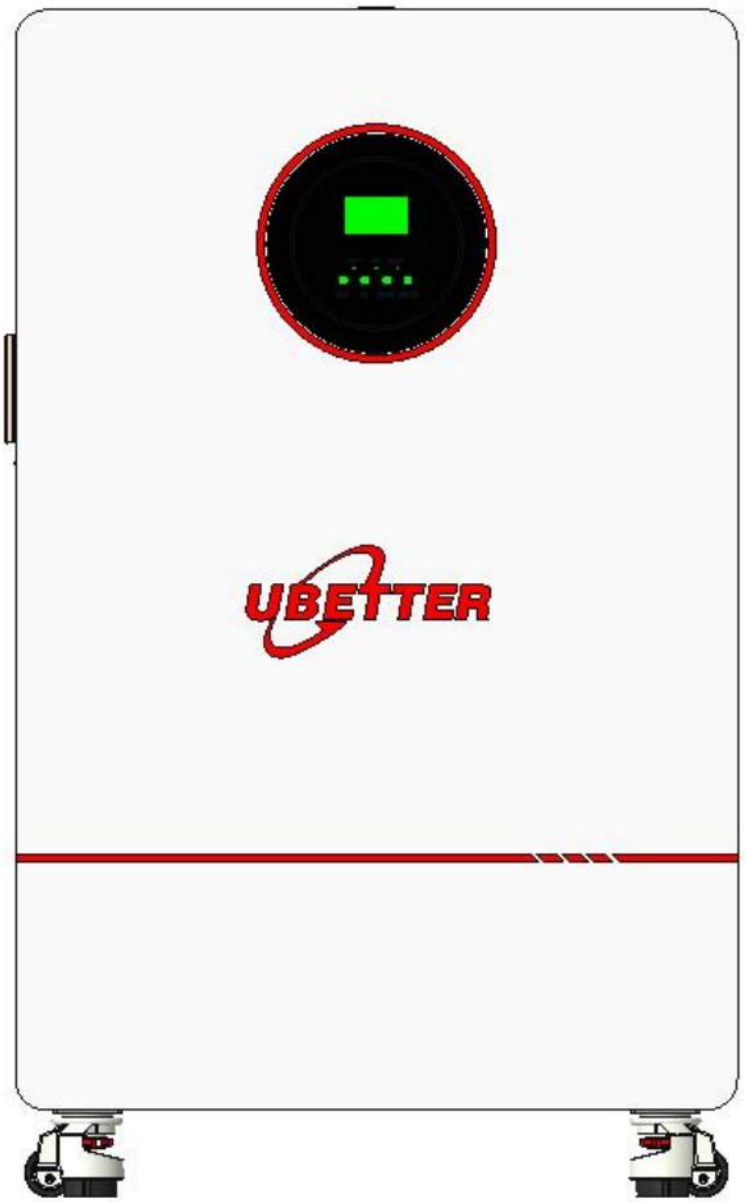
2.2 Battery pack specifications

	No.	project	Common parameters	remarks
battery pack	1	combinatorial method	15S1P	
	2	rated capacity	314Ah	Standard discharge followed by standard charge (packaging)
	3	working voltage	48V	Average operating voltage
	4	The voltage at the end of discharge	37.5V	Discharge cutoff voltage
	5	charging voltage	54.75V	Charging cut-off voltage
	6	Internal impedance	≤50mΩ	The internal resistance measured at 1KHZ AC must be measured after charging 50% using a new battery that has been shipped within one week and has fewer than 5 cycles.
	7	Standard charging	Constant current 40A Constant voltage see No.5 0.02CA cut off	Charging time: about 5 hours
	8	Maximum continuous charging current	100A	
	9	Standard discharge	Constant current: 20A The termination voltage is shown in No.4	
	10	Maximum continuous discharge current	100A	
	11	operating temperature range	Charging temperature range: 0~55℃	60±25% relative humidity.
			Discharge: -20~65℃	
	12	Storage temperature range	Less than 12 months: -10 to 35℃	60±25% relative humidity, in transport state.
			Less than 3 months: -10 to 35℃	
			Less than 7 days: -20~45℃	
	13	size	885*580*234mm	
	14	weight	Weight: approximately 120kg	

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3. Appearance and structural dimensions

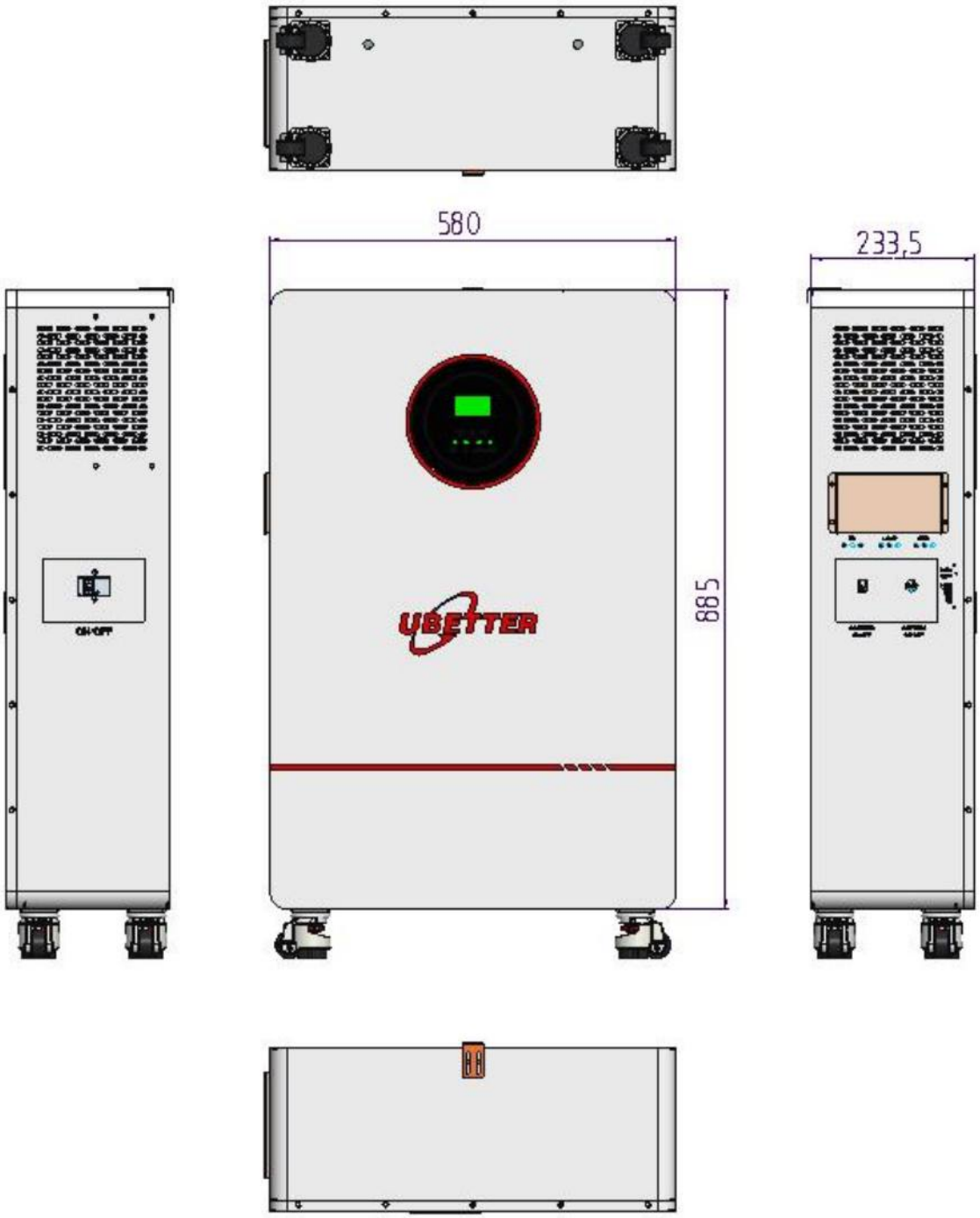
There shall be no scratches, burrs or other mechanical scratches and the connector shall be free of rust and dirt. See the drawings attached to the battery for structure and dimensions



L （length）	885mm	W （width）	580mm	H （altitude）	234mm
weight	120kg	Maximum charge capacity	100A	Maximum discharge	100A
Unit(mm)					

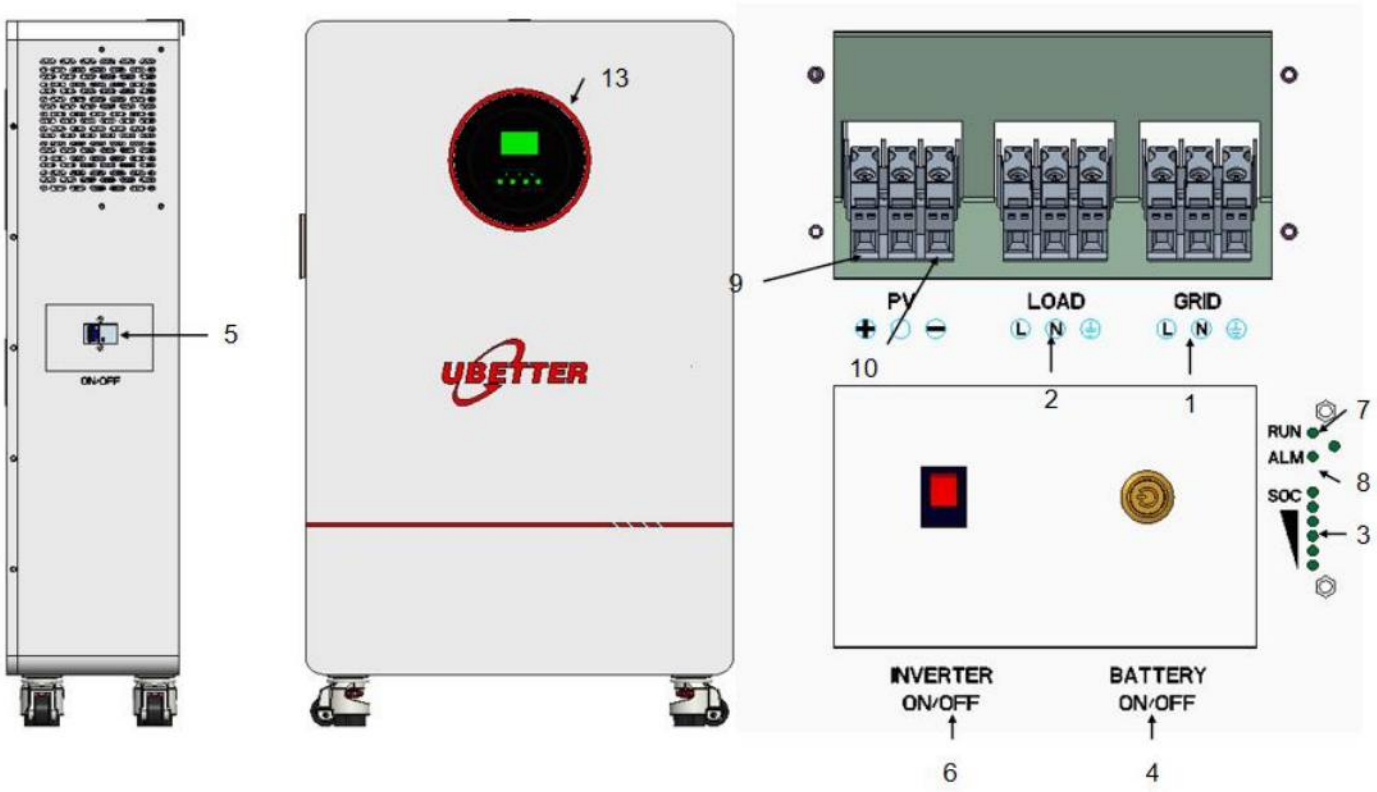
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48V-314Ah 15kWh LiFePO4 product size diagram



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4. The case structure of the battery Pack



No.	description	silk-screen	remarks
1	AC import	GRID	Output connector
2	AC output	LOAD	Output connector
3	battery status	SOC	Help users intuitively understand the remaining power
4	Battery switch	BAT/SW	Start/stop battery power
5	air switch	/	crowbar
6	Inverter switch	INVERTER/SW	Start/stop the inverter power supply
7	Operation indicator light	RUN	Operation status indicator
8	Alarm indicator	ALM	Fault alarm light
9	Photovoltaic input positive electrode	PV+	Photovoltaic input connector
10	Photovoltaic input negative electrode	PV-	Photovoltaic input connector

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6 battery test equipment

6.1 Dimension measuring instrument

Dimension measurement should be carried out with an instrument of no less than 0.1mm accuracy.

6.2 voltmeter

The standard class or more sensitive category specified in the national standard shall be used, and the internal resistance shall not be less than 10 kΩ/volt.

6.3 ammeter

The standard class or more sensitive category specified in the national standard should be used. The total external resistance including the ammeter and wire should be less than 0.01 ohms.

6.4 Impedance table

The impedance should be measured using the sinusoidal AC method (AC 1kHz LCR meter).

7. Standard test conditions

The battery shall undergo testing within one month after shipment from our factory, with the number of cycles prior to testing not exceeding five. Unless otherwise specified, all tests and measurements shall be conducted under conditions of 23±2℃ temperature, relative humidity below 75%, and air pressure ranging between 86Kpa and 106Kpa. Additionally, a 30-minute rest period is required for both charging and discharging processes, unless otherwise defined.

8. Storage and others

8.1 long-term storage

For long-term storage (unused for over three months), batteries should be kept in a dry and cool place. Maintain the battery's storage voltage at 48V-49.5V, storing it under conditions of 25±2℃ temperature and 45%-75% humidity. Batteries that remain unused for extended periods should be recharged every three months to ensure their voltage stays within the specified range.

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8.2 other

Any matter not covered by these specifications shall be settled by the client and UBETTER through negotiation.

9. Revision of this specification

This specification may be changed with prior notice from UBETTER.

10. Appendix

Precautions and guidelines for handling lithium ion rechargeable batteries

preface

This document, Handling Precautions and Guidelines for Lithium-ion Rechargeable Batteries, applies to battery units manufactured by UBETTER.

Note (1): If the customer needs to use other applications or operating conditions not described in this document, the customer should contact UBETTER in advance. Additional experiments may be required to verify performance and safety under these conditions.

Note (2): If the battery is used outside of the conditions described in this document, UBETTER will not be liable for any accident.

Note (3): If necessary, UBETTER will notify the customer in writing of improvements to the correct use and handling of batteries.

warning against danger！

-Do not immerse the battery in water or let it get wet.

-Do not use or store batteries near heat sources such as fire sources or heaters.

-Do not use any charger that is not recommended by UBETTER.

-Do not reverse the positive (+) and negative (-) terminals.

-Do not connect the battery directly to a wall outlet or car cigarette lighter outlet.

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-Do not put the battery in fire or heat it directly.

Do not short circuit the battery by connecting wires or other metal objects to the positive (+) and negative (-) terminals.

-Do not puncture the battery case with nails or other sharp objects, and do not hammer it open or step on it.

-Do not hit, throw or subject the battery to severe physical impact.

-Do not weld battery terminals directly.

-Do not disassemble or modify the battery in any way.

-Do not put batteries in a microwave oven or pressurized container.

-Do not use batteries with primary batteries (such as dry batteries) or batteries of different capacities, types or brands.

Do not use if the battery emits an odor, generates heat, changes color or shape, or appears abnormal in any way. If the battery is in use or charging, remove it from the device or charger and stop using it immediately.

warn !

Never store or use batteries in areas with extreme heat, such as direct sunlight on car windows during hot weather. Excessive heat can cause batteries to overheat, which may reduce performance and shorten their lifespan. If a battery leak occurs and electrolyte enters your eyes, avoid rubbing them. Instead, rinse immediately with clean running water and seek medical help right away. Left untreated, the leaked electrolyte could cause eye injuries.

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